

Charley M. Wu

Curriculum Vitae

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Nationality: Canadian

Academic Background

Current Position

- 2025 – **Professor (W3, tenured)**, *Chair of Computational Cognitive Science, Center for Cognitive Science, Institute of Psychology, Department of Human Sciences, TU Darmstadt, Darmstadt, Germany.*
- 2020 – 2025 **Independent Research Group Leader**, *Human and Machine Cognition Lab, University of Tübingen, Tübingen, Germany.*
Jointly funded by the Excellence Cluster “Machine Learning for Science” and the Tübingen AI center

Previous Positions

- 2019 – 2020 **Post-Doctoral Fellow**, *Department of Psychology, Harvard University, Cambridge, MA.*
Advised by Fiery A. Cushman and Samuel J. Gershman.
- 2019 **Post-Doctoral Fellow**, *Center for Adaptive Rationality (ARC), Max Planck Institute for Human Development, Berlin, Germany.*
- 2018 **Visiting Research Fellow**, *Computational Cognitive Neuroscience Lab; hosted by Samuel J. Gershman, Harvard University, Cambridge, MA.*
- 2016-2019 **Pre-Doctoral Fellow**, *Center for Adaptive Rationality (ARC) and Center for Adaptive Behavior and Cognition (ABC), Max Planck Institute for Human Development, Berlin, Germany.*

Education

- 2016-2019 **Dr. rer. nat. (Ph.D.) Psychology**, *Humboldt University of Berlin, Berlin, Germany, Summa Cum Laude; Advised by Björn Meder.*
- 2013-2015 **M.Sc. Cognitive Science**, *University of Vienna, Vienna, Austria, with Distinction.*
- 2004-2009 **B.A. Philosophy**, *University of British Columbia, Vancouver, Canada, Dean's List.*

Funding and Awards

- 2026 **LOEWE Start Professorship**, *Hessian Ministry for Science and the Arts, ~ €2 million over 6 years.*
- 2025 **ERC Starting Grant**, *European Research Council, C⁴: Compositional Compression in Cognition and Culture, ~ €1.5 million over 5 years.*
- 2024 **Rising Star**, *Association for Psychological Science (APS).*
- 2023 **Computational Summer School on Modeling Social and Collective Behavior (COS-MOS) Konstanz 2023**, *William K. and Katherine W. Estes Fund (with additional funding from Cluster of Excellence “Centre for the Advanced Study of Collective Behaviour” and the University of Konstanz), Co-Organizer: Wataru Toyokawa, ~ €25k.*

- 2023 **Exploring the Role of Approximate Causal Models in Human Decision-Making**, *Reinhard-Frank Stiftung*, Co-PI: Rahul Bhui (MIT), €27,860 .
- 2022 **Computational Summer School on Modeling Social and Collective Behavior (COS-MOS) Konstanz 2022**, *William K. and Katherine W. Estes Fund (with additional funding from Cluster of Excellence "Centre for the Advanced Study of Collective Behaviour")*, Co-Organizer: Wataru Toyokawa, ~ €25k.
- 2021 **Cumulative Culture in AI** (Postdoc funding), *Tübingen AI Center, supported by the Federal Ministry of Education and Research (BMBF)*, Co-PI: Claudio Tennie, ~ €121k.
- 2021 **Compositionality in Minds and Machines** (Mini-graduate School), *Innovation Fund Program of the Cluster of Excellence "Machine Learning: New Perspectives for Science"*, University of Tübingen, Co-PI: Martin Butz, ~ €114k.
- 2021 **Machine Learning for Education** (Mini-graduate School), *Innovation Fund Program of the Cluster of Excellence "Machine Learning: New Perspectives for Science"*, University of Tübingen, Co-PI: Álvaro Tejero-Cantero, ~ €114k.
- 2019 **Dean's Competitive Fund for Promising Research**, *Harvard University*, Cambridge, MA (written with and awarded to Sam Gershman), \$33,353 (USD).
- 2019 **Glushko and Samuelson Student Travel Grant**, *40th Annual Conference of the Cognitive Science Society*, Montreal, QC, \$500 (USD).
- 2016-2019 **Pre-Doctoral Fellowship**, *International Max Planck Research School on Adapting Behavior in a Fundamentally Uncertain World*, Joint PhD Fellowship in Psychology, Economics, and Law, ~€100k.
- 2011-2012 **Joseph-Armand Bombardier Canada Graduate Scholarship**, *Social Sciences and Humanities Research Council of Canada (SSHRC)*, Canada, \$17,500 (CAD), *Declined*.

Publications

In Prep

submitted Wong, E., Hauser, T. U., Chandrasekaran, A., Chandrasekaran, A., Pietrini, P., & **Wu**, C. M. (submitted). Uncertainty tolerance dissociates sensation seeking and impulsivity. *PsyArXiv*. doi:[10.31234/osf.io/yq8ce_v1](https://doi.org/10.31234/osf.io/yq8ce_v1)

Vélez, N., **Wu***, C. M., Gershman, S. J., & Schulz*, E. (submitted). The rise and fall of technological development in virtual communities. *PsyArxiv*. doi:[10.31234/osf.io/tz4dn](https://doi.org/10.31234/osf.io/tz4dn). (*Joint senior authorship.)

Ten, A., Sakaki, M., Breit, S., Chandrasekaran, A., Murayama, K., & **Wu**, C. M. (submitted). In search of lost memories: modeling exploration with forgetful generalization. *PsyArxiv*. doi:[0.31234/osf.io/hupq5_v1](https://doi.org/0.31234/osf.io/hupq5_v1)

Meder, B., Sterf, M., **Wu**, C. M., & Guggenmos, M. (submitted). Dopamine depletion in parkinson's increases directed but not random exploration. *PsyArxiv*. doi:[10.31234/osf.io/y2buv_v1](https://doi.org/10.31234/osf.io/y2buv_v1)

Goetmaeckers, F., **Wu**, C. M., Verguts, T., & Braem, S. (submitted). Environment-sensitive generalization and exploration strategies. *PsyArXiv*. doi:[10.31234/osf.io/35gyn_v1](https://doi.org/10.31234/osf.io/35gyn_v1)

Peer reviewed

- 2026 Palminteri, S. & **Wu**, C. M. (2026). Beyond computational functionalism: the behavioral inference principle for machine consciousness. *Neuroscience of Consciousness*, 2026(1), niag002. doi:[10.1093/nc/niag002](https://doi.org/10.1093/nc/niag002)
 - 2025 Zhou, H., Mantiuk, F., Nagy, D. G., & **Wu**, C. M. (2025). Agent-centric learning: from external reward maximization to internal knowledge curation. In *Reinforcement Learning Conference: Finding the Frame Workshop*. doi:[10.48550/arXiv.2507.22255](https://doi.org/10.48550/arXiv.2507.22255)
- Wu**, C. M., Meder, B., & Schulz, E. (2025). Unifying principles of generalization: past, present, and future. *Annual Reviews of Psychology*, 76, 275–302. doi:[10.1146/annurev-psych-021524-110810](https://doi.org/10.1146/annurev-psych-021524-110810)
- Wu**, C. M., Deffner, D., Kahl, B., Meder, B., Ho, M. K., & Kurvers, R. H. (2025). Adaptive mechanisms of social and asocial learning in immersive foraging environments. *Nature Communications*, 16, 3539. doi:[10.1038/s41467-025-58365-6](https://doi.org/10.1038/s41467-025-58365-6)
- Witt, A., Palminteri, S., & **Wu**, C. M. (2025). Flexibly biased learning rates in social learning. In A. Ruggeri, D. Barner, C. Walker, & N. Bramley (Eds.), *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. San Francisco, CA: Cognitive Science Society. doi:[10.31234/osf.io/bcrw9_v1](https://doi.org/10.31234/osf.io/bcrw9_v1)
- Szöcs, O., Zhou, H., & **Wu**, C. M. (2025). The forest for the trees: global vs. local advice in human-ai interaction. In A. Ruggeri, D. Barner, C. Walker, & N. Bramley (Eds.), *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. San Francisco, CA: Cognitive Science Society. doi:[10.31234/osf.io/ha6nr_v1](https://doi.org/10.31234/osf.io/ha6nr_v1)
- Orujlu, T., Gumbsch, C., Butz, M. V., & **Wu**, C. M. (2025). Reframing attention as a reinforcement learning problem for causal discovery. In *Reinforcement Learning Conference: Causal Reinforcement Learning Workshop*. doi:[10.48550/arXiv.2507.13920](https://doi.org/10.48550/arXiv.2507.13920)
- Nagy, D. G., Shen, T., Zhou, H., **Wu**, C. M., & Dayan, P. (2025). Analogy making as amortised model construction. In *Reinforcement Learning Conference: Finding the Frame Workshop*. doi:[10.48550/arXiv.2507.16511](https://doi.org/10.48550/arXiv.2507.16511)
- Nagy, D. G., Orban*, G., & **Wu***, C. M. (2025). Adaptive compression as a unifying framework for episodic and semantic memory. *Nature Reviews Psychology*. doi:[10.1038/s44159-025-00458-6](https://doi.org/10.1038/s44159-025-00458-6). (*Joint senior authorship.)
- Meder, B., **Wu**, C. M., & Rebitschek, F. G. (2025). Causal analysis of absolute and relative risk reductions. *Journal of Mathematical Psychology*. doi:[10.1016/j.jmp.2025.102942](https://doi.org/10.1016/j.jmp.2025.102942)
- Mantiuk, F., Zhou, H., & **Wu**, C. M. (2025). From curiosity to competence: how world models interact with the dynamics of exploration. In A. Ruggeri, D. Barner, C. Walker, & N. Bramley (Eds.), *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. San Francisco, CA: Cognitive Science Society. doi:[10.48550/arXiv.2507.08210](https://doi.org/10.48550/arXiv.2507.08210)
- Dettki, H., Lake, B. M., **Wu**, C. M., & Rehder, B. (2025). Do large language models reason causally like us? even better? In A. Ruggeri, D. Barner, C. Walker, & N. Bramley (Eds.), *Proceedings of the 47th Annual Conference of the Cognitive Science Society*. San Francisco, CA: Cognitive Science Society. doi:[10.48550/arXiv.2502.10215](https://doi.org/10.48550/arXiv.2502.10215)

- 2024 Zhou, H., Nagy, D. G., & **Wu**, C. M. (2024). Harmonizing program induction with rate-distortion theory. In *Proceedings of the 46th Annual Conference of the Cognitive Science Society*. Cognitive Science Society.
- Zhou, H., Bamler, R., **Wu***, C. M., & Tejero-Cantero*, A. (2024). Predictive, scalable and interpretable knowledge tracing on structured domains. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*. (*Joint senior authorship.)
- Wu**, C. M., Dale, R., & Hawkins, R. D. (2024). Group coordination catalyzes individual and cultural intelligence. *Open Mind*, 8, 1037–1057. doi:[10.1162/opmi_a_00155](https://doi.org/10.1162/opmi_a_00155)
- Wong, E., Hauser, T. U., Pietrini, P., & **Wu**, C. M. (2024). Shock to thrill: linking sensation and information seeking. In *Proceedings of the 46th Annual Conference of the Cognitive Science Society*. Cognitive Science Society. doi:[10.31234/osf.io/xa6yk](https://doi.org/10.31234/osf.io/xa6yk)
- Witt, A., Toyokawa, W., Lala, K. N., Gaissmaier, W., & **Wu**, C. M. (2024). Humans flexibly integrate social information despite interindividual differences in reward. *Proceedings of the National Academy of Sciences*, 121(39), e2404928121. doi:[10.1073/pnas.2404928121](https://doi.org/10.1073/pnas.2404928121)
- Nanni-Zepeda, M., DeGutis, J., **Wu**, C. M., Rothlein, D., Fan, Y., Grimm, S., . . . Zuberer, A. (2024). Neural signatures of shared subjective affective engagement and disengagement during movie viewing. *Human Brain Mapping*, 45(4), e26622. doi:[10.1002/hbm.26622](https://doi.org/10.1002/hbm.26622)
- Hamidi, M., Khajehabdollahi, S., Giannakakis, E., Schäfer, T., Levina, A., & **Wu**, C. M. (2024). Modular growth of hierarchical networks: efficient, general, and robust curriculum learning. In *Proceedings of the 2024 Artificial Life Conference (ALIFE)*. doi:[10.48550/arXiv.2406.06262](https://doi.org/10.48550/arXiv.2406.06262)
- Deffner, D., Mezey, D., Kahl, B., Schakowski, A., Romanczuk, P., **Wu**, C. M., & Kurvers, R. (2024). Collective incentives reduce over-exploitation of social information in unconstrained human groups. *Nature Communications*. doi:[10.1038/s41467-024-47010-3](https://doi.org/10.1038/s41467-024-47010-3)
- Collins, R. N., Mandel, D. R., Karvetski, C. W., **Wu**, C. M., & Nelson, J. D. (2024). The wisdom of the coherent: improving correspondence with coherence-weighted aggregation. *Decision*, 11, 60–85. doi:[10.1037/dec0000211](https://doi.org/10.1037/dec0000211)
- 2023 Zhou, H., Tejero-Cantero*, Á., & **Wu***, C. M. (2023). The dynamic and structured nature of learning and memory. In L. Hunt, C. Summerfield, T. Konkle, E. Fedorenko, & T. Naselaris (Eds.), *Proceedings of the 2023 Conference on Cognitive Computational Neuroscience*. Oxford, UK. (*Joint senior authorship.)
- Xiong, Y., Moneta, N., Bányaí, M., & **Wu**, C. M. (2023). Selective memory for reward-relevant features is modulated by expertise during reward learning. In L. Hunt, C. Summerfield, T. Konkle, E. Fedorenko, & T. Naselaris (Eds.), *Proceedings of the 2023 Conference on Cognitive Computational Neuroscience*. Oxford, UK.
- Witt, A., Vasama, J., Vélez, N., & **Wu**, C. M. (2023). Playing to win or playing to learn? human performance in a social card game task. In L. Hunt, C. Summerfield, T. Konkle, E. Fedorenko, & T. Naselaris (Eds.), *Proceedings of the 2023 Conference on Cognitive Computational Neuroscience*. Oxford, UK.

- Witt, A., Toyokawa, W., Lala, K., Gaissmaier, W., & **Wu**, C. M. (2023). Social learning with a grain of salt. In M. Goldwater, F. Anggoro, B. Hayes, & D. Ong (Eds.), *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Sydney, Australia: Cognitive Science Society. doi:[10.31234/osf.io/c3fuq](https://doi.org/10.31234/osf.io/c3fuq)
- Uchiyama, R., Tennie, C., & **Wu**, C. M. (2023). Model-based assimilation transmits and recombines world models. In L. Hunt, C. Summerfield, T. Konkle, E. Fedorenko, & T. Naselaris (Eds.), *Proceedings of the 2023 Conference on Cognitive Computational Neuroscience*. Oxford, UK. doi:[10.31234/osf.io/v69jy](https://doi.org/10.31234/osf.io/v69jy)
- Smith, A. L., Heuschkel, S., Keplinger, K., & **Wu**, C. M. (2023). Constructing and deconstructing bias: modeling privilege and mentorship in agent-based simulations. In L. Hunt, C. Summerfield, T. Konkle, E. Fedorenko, & T. Naselaris (Eds.), *Proceedings of the 2023 Conference on Cognitive Computational Neuroscience*. Oxford, UK. doi:[10.48550/arXiv.2304.02351](https://doi.org/10.48550/arXiv.2304.02351)
- Rubino, V., Hamidi, M., Dayan, P., & **Wu**, C. M. (2023). Compositionality under time pressure. In M. Goldwater, F. Anggoro, B. Hayes, & D. Ong (Eds.), *Proceedings of the 45th Annual Conference of the Cognitive Science Society*. Sydney, Australia: Cognitive Science Society. doi:[10.31234/osf.io/z2648](https://doi.org/10.31234/osf.io/z2648)
- Rubino, V., Dayan, P., & **Wu**, C. M. (2023). Biases towards compositionally simpler hypotheses are robust and unaffected by learning. In L. Hunt, C. Summerfield, T. Konkle, E. Fedorenko, & T. Naselaris (Eds.), *Proceedings of the 2023 Conference on Cognitive Computational Neuroscience*. Oxford, UK.
- Moneta, N., **Wu**, C. M., Doeller, C. F., & Schuck, N. W. (2023). Reward morphs non-spatial cognitive maps in humans. In L. Hunt, C. Summerfield, T. Konkle, E. Fedorenko, & T. Naselaris (Eds.), *Proceedings of the 2023 Conference on Cognitive Computational Neuroscience*. Oxford, UK.
- Haridi, S., **Wu**, C. M., Dasgupta, I., & Schulz, E. (2023). The scaling of mental computation in a sorting task. *Cognition*, 241, 105605. doi:[10.1016/j.cognition.2023.105605](https://doi.org/10.1016/j.cognition.2023.105605)
- Hamidi, M., Bányai, M., & **Wu**, C. M. (2023). One bottleneck is not enough. In L. Hunt, C. Summerfield, T. Konkle, E. Fedorenko, & T. Naselaris (Eds.), *Proceedings of the 2023 Conference on Cognitive Computational Neuroscience*. Oxford, UK.
- Giron, A. P., Ciranka, S., Schulz, E., van den Bos, W., Ruggeri, A., Meder, B., & **Wu**, C. M. (2023). Developmental changes in exploration resemble stochastic optimization. *Nature Human Behaviour*. doi:[10.1038/s41562-023-01662-1](https://doi.org/10.1038/s41562-023-01662-1)
- Brändle, F., **Wu**, C. M., & Schulz, E. (2023). Learning progress and uncompensated rewards as motivational drivers of engagement. In L. Hunt, C. Summerfield, T. Konkle, E. Fedorenko, & T. Naselaris (Eds.), *Proceedings of the 2023 Conference on Cognitive Computational Neuroscience*. Oxford, UK.
- 2022 **Wu**, C. M., Vélez, N., & Cushman, F. A. (2022). Representational exchange in human social learning: Balancing efficiency and flexibility. In I. C. Dezza, E. Schulz, & C. M. Wu (Eds.), *The Drive for Knowledge: The Science of Human Information-Seeking*. Cambridge: Cambridge University Press.

- Wu**, C. M., Schulz, E., Pleskac, T. J., & Speekenbrink, M. (2022). Time pressure changes how people explore and respond to uncertainty. *Scientific Reports*, 12, 1–14. doi:<https://doi.org/10.1038/s41598-022-07901-1>
- Vélez, N., **Wu**, C. M., & Cushman, F. A. (2022). Representational exchange in social learning: blurring the lines between the ritual and instrumental. *Behavioral and Brain Sciences*, 45, e271. doi:[10.1017/S0140525X22001339](https://doi.org/10.1017/S0140525X22001339)
- Ludwig, T., **Wu**, C. M., & Schulz, E. (2022). Connecting exploration, generalization, and planning in correlated trees. In *Proceedings of the 44rd Annual Conference of the Cognitive Science Society*. Toronto, Canada: Cognitive Science Society.
- Dezza, I. C., Schulz, E., & **Wu**, C. M. (Eds.). (2022). *The Drive for Knowledge: The Science of Human Information-Seeking*. Cambridge: Cambridge University Press. doi:<https://doi.org/10.1017/9781009026949>
- Ciranka, S., Linde-Domingo, J., Padezhki, I., Wicharz, C., **Wu**, C. M., & Spitzer, B. (2022). Asymmetric learning facilitates human inference of transitive relations. *Nature Human Behaviour*. doi:<https://doi.org/10.1038/s41562-021-01263-w>
- 2021 Zuberer, A., Kucyi, A., Yamashita, A., **Wu**, C. M., Walter, M., Valera, E. M., & Esterman, M. (2021). Integration and segregation across large-scale intrinsic brain networks as a marker of sustained attention and task-unrelated thought. *NeuroImage*, 229, 117610. doi:[10.1016/j.neuroimage.2020.117610](https://doi.org/10.1016/j.neuroimage.2020.117610)
- Wu**, C. M., Schulz, E., & Gershman, S. J. (2021). Inference and search on graph-structured spaces. *Computational Brain & Behavior*, 125–147. doi:[10.1007/s42113-020-00091-x](https://doi.org/10.1007/s42113-020-00091-x)
- Wu**, C. M., Ho, M. K., Kahl, B., Leuker, C., Meder, B., & Kurvers, R. H. (2021). Specialization and selective social attention establishes the balance between individual and social learning. In T. Fitch, C. Lamm, H. Leder, & K. Teßmar-Raible (Eds.), *Proceedings of the 43rd Annual Conference of the Cognitive Science Society* (pp. 1921–1927). Vienna, Austria: Cognitive Science Society. doi:[10.1101/2021.02.03.429553](https://doi.org/10.1101/2021.02.03.429553)
- Meder, B., **Wu**, C. M., Schulz, E., & Ruggeri, A. (2021). Development of directed and random exploration in children. *Developmental Science*, e13095. doi:[10.1111/desc.13095](https://doi.org/10.1111/desc.13095)
- Humaidan, D., Otte, S., Gumbsch, C., **Wu**, C. M., & Butz, M. V. (2021). Latent event-predictive encodings through counterfactual regularization. In T. Fitch, C. Lamm, H. Leder, & K. Teßmar-Raible (Eds.), *Proceedings of the 43rd Annual Conference of the Cognitive Science Society* (pp. 1726–1731). Vienna, Austria: Cognitive Science Society. eprint: [2105.05894](https://arxiv.org/abs/2105.05894)
- 2020 **Wu**, C. M., Schulz, E., Garvert, M. M., Meder, B., & Schuck, N. W. (2020). Similarities and differences in spatial and non-spatial cognitive maps. *PLOS Computational Biology*, 16, 1–28. doi:[10.1371/journal.pcbi.1008149](https://doi.org/10.1371/journal.pcbi.1008149)
- Brändle, F., **Wu**, C. M., & Schulz, E. (2020). What are we curious about? *Trends in Cognitive Science*. doi:[10.1016/j.tics.2020.05.010](https://doi.org/10.1016/j.tics.2020.05.010)

- 2019 **Wu**, C. M., Schulz, E., & Gershman, S. J. (2019a). Generalization as diffusion: human function learning on graphs. In A. Goel, C. Seifert, & C. Freksa (Eds.), *Proceedings of the 41st Annual Conference of the Cognitive Science Society* (pp. 3122–3128). Montreal, QB: Cognitive Science Society.
- Wu**, C. M., Schulz, E., & Gershman, S. J. (2019b). Searching for rewards in graph-structured spaces. In *Proceedings of the 2019 Conference on Cognitive Computational Neuroscience*. doi:[10.32470/CCN.2019.1041-0](https://doi.org/10.32470/CCN.2019.1041-0)
- Wu***, C. M., Schulz*, E., Gerbaulet*, K., Pleskac, T. J., & Speekenbrink, M. (2019). Under pressure: The influence of time limits on human exploration. In A. Goel, C. Seifert, & C. Freksa (Eds.), *Proceedings of the 41st Annual Conference of the Cognitive Science Society* (pp. 1219–1225). Montreal, QB: Cognitive Science Society. (*Joint first authorship.)
- Tump*, A. N., **Wu***, C. M., Bouhlel, I., & Goldstone, R. L. (2019). The evolutionary dynamics of cooperation in collective search. In A. Goel, C. Seifert, & C. Freksa (Eds.), *Proceedings of the 41st Annual Conference of the Cognitive Science Society* (pp. 883–889). Montreal, QB: Cognitive Science Society. (*Joint first authorship.)
- Schulz, E., **Wu**, C. M., Ruggeri, A., & Meder, B. (2019). Searching for rewards like a child means less generalization and more directed exploration. *Psychological Science*, 30(11), 1561–1572. doi:[10.1177/0956797619863663](https://doi.org/10.1177/0956797619863663)
- Analytis, P. P., **Wu**, C. M., & Gelastopoulos, A. (2019). Make-or-break: chasing risky goals or settling for safe rewards? *Cognitive Science*, 43, e12743. doi:[10.1111/cogs.12743](https://doi.org/10.1111/cogs.12743)
- 2018 **Wu**, C. M., Schulz, E., Speekenbrink, M., Nelson, J. D., & Meder, B. (2018a). Generalization guides human exploration in vast decision spaces. *Nature Human Behaviour*, 2, 915–924. doi:[10.1038/s41562-018-0467-4](https://doi.org/10.1038/s41562-018-0467-4)
- Wu**, C. M., Schulz, E., Garvert, M. M., Meder, B., & Schuck, N. W. (2018b). Connecting conceptual and spatial search via a model of generalization. In T. T. Rogers, M. Rau, X. Zhu, & C. W. Kalish (Eds.), *Proceedings of the 40th Annual Conference of the Cognitive Science Society* (pp. 1183–1188). Austin, TX: Cognitive Science Society.
- Schulz, E., **Wu**, C. M., Huys, Q. J., Krause, A., & Speekenbrink, M. (2018). Generalization and search in risky environments. *Cognitive Science*, 42, 2592–2620. doi:[10.1111/cogs.12695](https://doi.org/10.1111/cogs.12695)
- Bouhlel*, I., **Wu***, C. M., Hanaki, N., & Goldstone, R. L. (2018). Sharing is not erring: pseudo-reciprocity in collective search. In T. T. Rogers, M. Rau, X. Zhu, & C. W. Kalish (Eds.), *Proceedings of the 40th Annual Conference of the Cognitive Science Society* (pp. 156–161). Austin, TX: Cognitive Science Society. (*Joint first authorship.)
- 2017 **Wu**, C. M., Schulz, E., Speekenbrink, M., Nelson, J. D., & Meder, B. (2017). Mapping the unknown: the spatially correlated multi-armed bandit. In G. Gunzelmann, A. Howes, T. Tenbrink, & E. J. Davelaar (Eds.), *Proceedings of the 39th Annual Meeting of the Cognitive Science Society* (pp. 1357–1362). Austin, TX: Cognitive Science Society.
- Wu**, C. M., Meder, B., Filimon, F., & Nelson, J. D. (2017). Asking better questions: how presentation formats influence information search. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 8, 1274–1297. doi:[doi:10.1037/xlm0000374](https://doi.org/10.1037/xlm0000374)

- 2016 Barkoczi, D., Analytis, P. P., & Wu, C. M. (2016). Collective search on rugged landscapes: a crossenvironmental analysis. In A. Papafragou, D. Grodner, D. Mirman, & J. Trueswell (Eds.), *Proceedings of the 38th Annual Conference of the Cognitive Science Society* (pp. 918–923). Austin, TX: Cognitive Science Society.

Teaching

- 2023-25 **General Principles of Human and Machine Learning**, *Lecture: Cognitive Science/Graduate Training Center for Neuroscience/Psychology/Computer Science*, University of Tübingen, [\[Course website\]](#).
- 2022-2023 **Cognitive maps and model-based reinforcement learning**, *Seminar: Cognitive Science/Graduate Training Center for Neuroscience*, University of Tübingen, [\[Course website\]](#).
- 2022-2023 **Tutorials on Modeling Social Learning and Collective Behavior**, *The Computational Summer school on Modeling Social and collective behavior (COSMOS)*, Konstanz, DE, [\[Code notebooks\]](#).
- 2022-2023 **Scientific reasoning: Crafting research questions and arguments**, *Computation and Cognition Tübingen Summer Internship (CaCTüS) workshop*, Max Planck Institute for Biological Cybernetics, Tübingen, Germany.
- 2021 **Generalization in Reinforcement Learning**, *Guest lecture: Introduction to Cognitive Psychology*, University of Ghent, Ghent, Belgium (via Zoom) [\[Slides\]](#).
- 2020 **Computational Modeling**, *Graduate student workshop*, Max Planck Institute for Biological Cybernetics, Tübingen, Germany (via Zoom).
- 2019-2020 **Representation Learning in Reinforcement Learning Seminar**, *Co-organizer and regular Speaker*, Harvard University, Center for Brain Science, Cambridge, MA. [\[Notes\]](#).
- 2015-2019 **Berlin Machine Learning Seminar**, *Regular Speaker*, Berlin, Germany.
- 2018 **Introduction to Computational Modeling**, *Graduate and undergraduate workshop*, MPRG: iSearch Research Retreat, Bensdorf, Germany.
- 2018 **Intro to Cognitive Modeling**, Max Planck Institute for Human Development, Berlin, Germany. (Teaching Assistant to Prof. Björn Meder).
- 2017 **Computational Models of Cognition**, *Lecture*, Berlin School of Mind and Brain (PhD Program), Humboldt University, Berlin, Germany.
- 2016-2017 **Math and Methods Tutorial Series**, *Organizer and regular speaker*, Center for Adaptive Behavior and Cognition (ABC), Berlin, Germany.

Supervision

PhD Theses

- Alexandra Witt**. Department of Computer Science, University of Tübingen. PhD Thesis: *The Neural and Behavioral Basis for Social Learning and Theory of Mind Inference*. (2021-2025)
- Turan Orujlu**. Department of Computer Science, University of Tübingen. PhD Thesis: *Curriculum learning for self-directed goal discovery in physical environments* (2022-Present)
- Andria Smith**. Department of Computer Science, University of Stuttgart. PhD Thesis: *Computational basis of organizational diversity and leadership* (2022-Present)
- Hanqi Zhou**. Department of Computer Science, University of Tübingen. PhD Thesis: *Assisting Adaptive Human Learning in Structured Domains* (2022-Present)
- Mani Hamidi**. Department of Computer Science, University of Tübingen. PhD Thesis: *Keeping it Systematically Simple: Heuristics for the control of representation complexity over the course of*

learning (2021-Present)

Rachel Burns. Mathematics and Natural Sciences Faculty, University of Tübingen. PhD Thesis: *Creating a Haptic Empathetic Robot Animal That Feels Touch and Emotion* (2024). External examiner.

Jan-Philipp Fränken. College of Arts, Humanities and Social Sciences, University of Edinburgh. PhD Thesis: *Reasoning about quantities and concepts: studies in social learning* (2022). External examiner.

Master's and Bachelor's theses

Fryderyk Mantiuk. Department of Computer Science, University of Tübingen. Masters Thesis: *Using Large Language Models for Program Induction* (ongoing)

Pulkit Goyal. Graduate Training Center for Neuroscience, University of Tübingen. Masters Thesis: *Building Visual Semantic Bias in Curious Exploration during Free Play* (ongoing)

Anna-Lena von Behren. Department of Computer Science, University of Tübingen. Masters Thesis: *Improving 3D Gaze Estimation in Virtual Reality using Depth Data* (ongoing)

Orsolya Szöcs. Department of Computer Science, University of Tübingen. Masters Thesis: *Do humans prefer global or local AI advice?* (ongoing)

Anna-Lena von Behren. Department of Computer Science, University of Tübingen. Masters Thesis: *Improving 3D Gaze Estimation in Virtual Reality using Depth Data* (ongoing)

Simon Heuschel. Department of Cognitive Science, University of Tübingen. Masters Thesis: *An Agent-Based Model Framework to Understand Inequality and Diversity Management in Groups* (2024)

Sahiti Chebolu. Graduate Training Center for Neuroscience, University of Tübingen. Masters Thesis: *Procrastination as temporal decision making* (2023)

Joel Vasama. Graduate Training Center for Neuroscience, University of Tübingen. Masters Thesis: *Testing for Adaptive Use of Social Learning Mechanisms* (2023)

Marcel De Sutter. Department of Cognitive Science, University of Tübingen. Masters Thesis: *Physical and psychological reasoning in artificial cognitive systems: A benchmark study of Loci's object vs. agent recognition* (2023)

Sebastian Breit. Department of Cognitive Science, University of Tübingen. Masters Thesis: *In search of lost memories: modeling forgetful generalization* (2022-2023)

Anna Giron. Department of Cognitive Science, University of Tübingen. Masters Thesis: *The Trajectory of Learning and Exploration Over the Lifespan* (2020-2021)

Theresa Horn. Department of Cognitive Science, University of Tübingen. Bachelors Thesis: *Use of Visual and Spatial Information in Human Search Behaviour* (2021)

Kimberly Gerbaulet. Institute of Cognitive Science, University of Osnabrück. Masters Thesis: *Under pressure: the effect of time pressure on directed and random exploration.* (2018-2019)

Thesis Advisory Committee

Constantin Ruhdorfer. PhD student. IMPRS IS, University of Stuttgart. (2024 - present)

Alexander Platt. PhD student. Max Planck School of Cognition, University of Tübingen. (2023 - present)

Maximilian Mittenbühler. PhD student. Cognitive Modeling group, University of Tübingen. (2022 - present)

Gabriela Iawama. PhD student. Hertie Institute for Clinical Brain Sciences., University of Tübingen. (2022 - present)

Ori Press. PhD student. Computational Vision and Neuroscience, University of Tübingen. (2022 - 2025)

Manuel Traub. PhD student. Cognitive Modeling group, University of Tübingen. (2021 - present)

Ruiqi He. PhD student. Rationality Enhancement Group, Max Planck Institute for Intelligent Systems. (2020 - 2025)

Lovis Hendrich. PhD student. Rationality Enhancement Group, Max Planck Institute for Intelligent Systems. (2020 - present)

Christian Gumbsch. PhD student. Autonomous learning, Max Planck Institute for Intelligent Systems. (2019 - 2024)

Professional Service

Associate Editor. Open Mind, MIT Press (2022-Present)

Grant Committee Member French National Research Agency (ANR) (2024-2025)

Steering Committee Member of the Cluster of Excellence – Machine Learning for Science, University of Tübingen (2021-2023)

Co-Organizer of The Computational Summer school on Modeling Social and collective behavior (COSMOS) 2022-2023, Konstanz, DE

Neuromatch mentor (2021)

Organizer of Cognition, Collectives, and Human Culture Workshop (part of CogSci 2020), Toronto, Canada (2020)

Organizer of the 17th annual Summer Institute on Bounded Rationality, Berlin, Germany (2018)

PhD Representative for the Max Planck Institute for Human Development (2017-2018)

Grant Reviewer National Science Foundation (NSF), German Research Foundation (DFG), and Human Frontier Science Program (HFSP)

Ad-hoc Reviewer for Nature Communications, Nature Human Behaviour, eLife, PLOS Computational Biology, Scientific Reports, Cognition, Cognitive Psychology, NeurIPS, Artificial Intelligence, JEP: G, JEP: LMC, Computational Brain and Behavior, Child Development, Cognitive, Affective, and Behavioral Neuroscience, Cognitive Science, Cognitive Computational Neuroscience, Cognitive Science Society, Mind & Society, and Futures & Foresight Science

Technical Skills

Programming Languages: Python, R, Matlab, Julia, JavaScript, HTML, PHP, CSS, and $\LaTeX$.

Languages: English (Native), German (C1), Chinese (Mandarin/Shanghainese; mother tongue), French (B1), Spanish (B1), and Russian (A1)